

Analysis PhD Exam, May 1997

Be sure to write all steps in a neat and logical fashion to receive credit. Give reasons for your steps.

Definition: A σ -algebra $\mathcal{F}$ is separable if $\mathcal{F} = \sigma\{A_j : j \geq 1\}$ for some sequence of sets A_j .

1. Let $\{\mathcal{F}_k\}$ be a sequence of separable σ -algebras of subsets of a set X . Let $\mathcal{F}$ be the smallest σ -algebra of subsets of X which contains $\mathcal{F}_k$ for each k . Prove that $\mathcal{F}$ is separable. [Hint: Use the monotone class theorem.]
2. Let $(X, \mathcal{F}, \mu)$ be a finite measure space and let (Y, Σ) be a measurable space. Let $f : X \rightarrow Y$ be $\mathcal{F}, \Sigma$ -measurable. Define $\nu(B) = \mu(f^{-1}(B))$ for $B \in \Sigma$. Show ν is a measure on Σ , and if $g : Y \rightarrow [0, \infty]$ is Σ -measurable, then $\int_Y g d\nu = \int_X g \circ f d\mu$.
3. Let m be Lebesgue measure on $\mathbb{R}$. Let $f_m = -1_{A_m}$, where $A_m = (m, \infty)$. Show (f_m) is increasing, but the monotone convergence theorem is not true. Why is this?
4. Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{J}, \nu)$ be two finite measure spaces. Construct the product measure $\mu \times \nu$ on $\mathcal{S} \otimes \mathcal{J}$. Be sure to prove $\mu \times \nu$ is countably additive.
5. Let (X, Σ, μ) be a finite measure space. Prove that the dual of $L^1(X, \Sigma, \mu)$ is isometrically isomorphic to $L^\infty(X, \Sigma, \mu)$.
6. Let $(X, \mathcal{F}, \mu)$ be a finite measure space. Suppose that $\mathcal{F}$ is separable. Show $L^1(X, \mathcal{F}, \mu)$ is separable.
7. Let $(X, \mathcal{F}, \mu)$ be a measure space. Let $\{f_m\}$ be a sequence of measurable functions such that $\int_X |f_m| d\mu < \frac{1}{2^m}$. Does it follow that $\sum_{m=1}^\infty f_m$ converges a.e. μ on X ?
8. Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \rightarrow \mathbb{R}$ be an integrable function. Let $F(x) = P(X < x)$, for $x \in \mathbb{R}$.
 - (a) Show $F : \mathbb{R} \rightarrow [0, 1]$ is monotone, left continuous, $\lim_{x \rightarrow -\infty} F(x) = 0$, and $\lim_{x \rightarrow \infty} F(x) = 1$.
 - (b) Show $\int_\Omega X^2 dP = \int_\mathbb{R} x^2 dF(x)$.
9. Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let (X_m) be a sequence of independent random variables on Ω .
 - (a) Show $(\overline{\lim} X_m > 3)$ is a tail event relative to (X_m) .
 - (b) If $P(\overline{\lim} X_m > 3) > 0.2$, what is $P(\overline{\lim} X_m > 3)$?
10. Let ℓ^∞ be the Banach space consisting of all real sequences (a_i) such that $\sup_i |a_i| < \infty$, with $\|(a_i)\| := \sup_i |a_i|$. Let c be the subspace of ℓ^∞ consisting of (a_i) such that $\lim a_i$ exists. Prove that there exists a continuous linear functional $F : \ell^\infty \rightarrow \mathbb{R}$ such that $F((a_i)) = \lim a_i$ for each $(a_i) \in c$. [Hint: Use the Hahn–Banach theorem.]